Yuri van der Burg

Generative AI for Image Reconstruction in Intensity Interferometry: a First Attempt

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Presenting at

a Meeting Organised by ....

Objective: High Resolution of Stellar Objects...

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Limitation: There Size of Aperture of Telescope...

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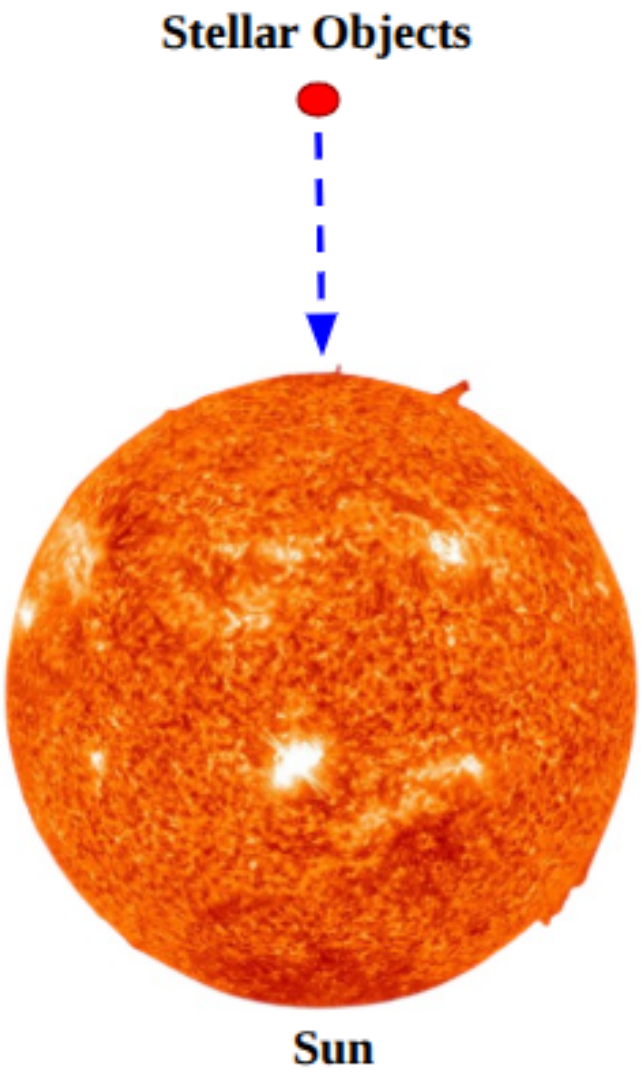
Challenges: The Extraction of Lost Quantities for high resolution...

Requirements of High Resolution

- Stellar objects appear almost like point sources.
- Angular sizes are extremely small ($\sim$ milliarcsecond or less).
- High angular resolution is required to resolve stellar surfaces.
- Understanding stellar structure is crucial for stellar evolution.
- The Sun is the only star currently resolved in detail.



Maximum Resolution with a Telescope

Show the diffraction formula and a figure (resolved with DOT)

The Intensity Interferometry

Profit and Loss

The Role of GAN

Image Reconstruction: The Result

The Next Steps.....

Thank You!!